Lecture 20: Section 5.5

The Definite Integral

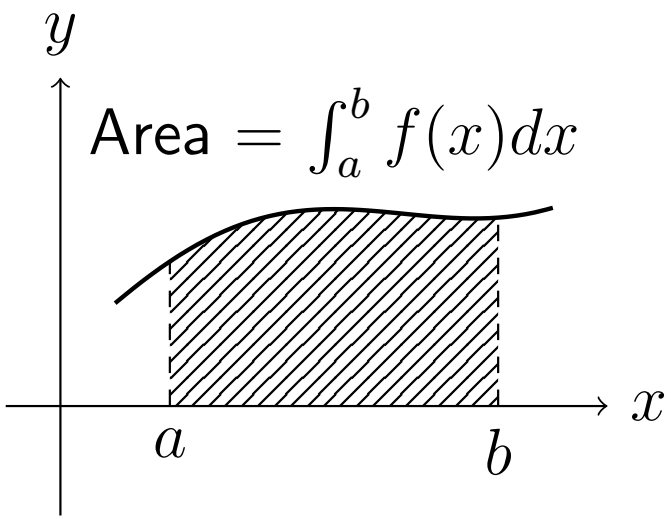
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What is the Definite Integral?

Definition

The **definite integral** of a function $f(x)$ from $x = a$ to $x = b$, written as $\int_a^b f(x)\,dx$, represents the *net signed area* between the graph of $f(x)$, the x-axis, and the vertical lines $x = a$ and $x = b$.



Key Idea: "Net signed" means area above the x-axis counts as positive, area below counts as negative.

Basic Algebraic Properties

These properties make calculations easier and are based on area intuition.

Property (Reversing Limits)

$$\int_a^b f(x)\,dx = -\int_b^a f(x)\,dx$$

Property (Same Upper and Lower Limit)

$$\int_a^a f(x)\,dx = 0$$

Property (Constant Multiple)

$$\int_a^b c \cdot f(x)\,dx = c\int_a^b f(x)\,dx \text{ for any constant } c.$$

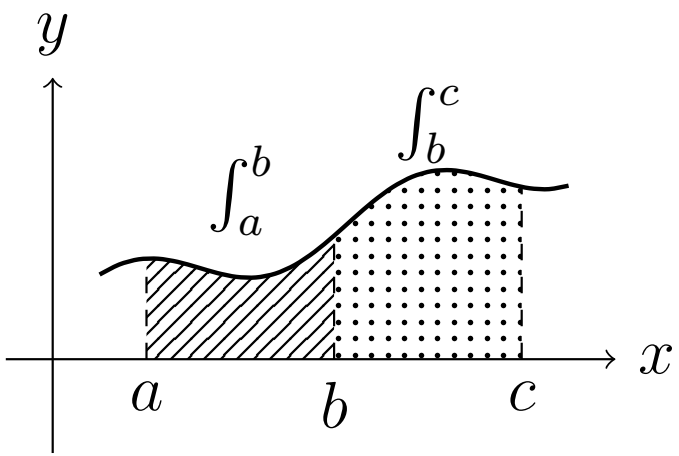
More Algebraic Properties

Property (Sum/Difference)

$$\int_a^b [f(x) \pm g(x)]\,dx = \int_a^b f(x)\,dx \pm \int_a^b g(x)\,dx$$

Property (Additivity of Interval)

$$\int_a^b f(x)\,dx + \int_b^c f(x)\,dx = \int_a^c f(x)\,dx$$



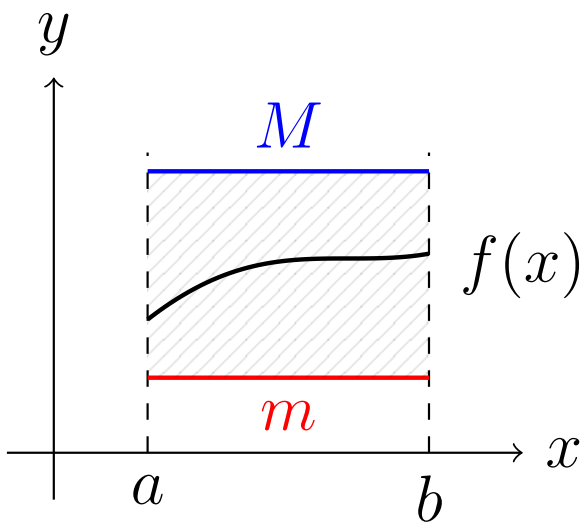
Property (Preservation of Inequality)

If $f(x) \geq g(x)$ *for all* x *in* $[a, b]$, *then* $\int_a^b f(x)\,dx \geq \int_a^b g(x)\,dx$.

Property (Bounds)

If $m \leq f(x) \leq M$ *for all* x *in* $[a, b]$, *then*

$$m(b-a) \leq \int_a^b f(x)\,dx \leq M(b-a).$$



Worked Examples: Using Properties

Example

Given $\int_1^5 f(x)dx = 7$ and $\int_1^5 g(x)dx = 3$, find $\int_1^5 [2f(x) - 5g(x)]dx$.

Solution:

$$\int_1^5 [2f(x) - 5g(x)]dx = \int_1^5 2f(x)dx - \int_1^5 5g(x)dx \quad \text{(Sum rule)}$$

$$= 2\int_1^5 f(x)dx - 5\int_1^5 g(x)dx \quad \text{(Const. multi.)}$$

$$= 2(7) - 5(3) = 14 - 15 = -1.$$

More Worked Examples

Example

Given $\int_2^8 f(x)dx = 10$ and $\int_5^8 f(x)dx = 3$, find $\int_2^5 f(x)dx$.

Solution: By additivity of the interval:

$$\int_2^8 f(x)dx = \int_2^5 f(x)dx + \int_5^8 f(x)dx$$

$$10 = \int_2^5 f(x)dx + 3$$

$$\int_2^5 f(x)dx = 7.$$

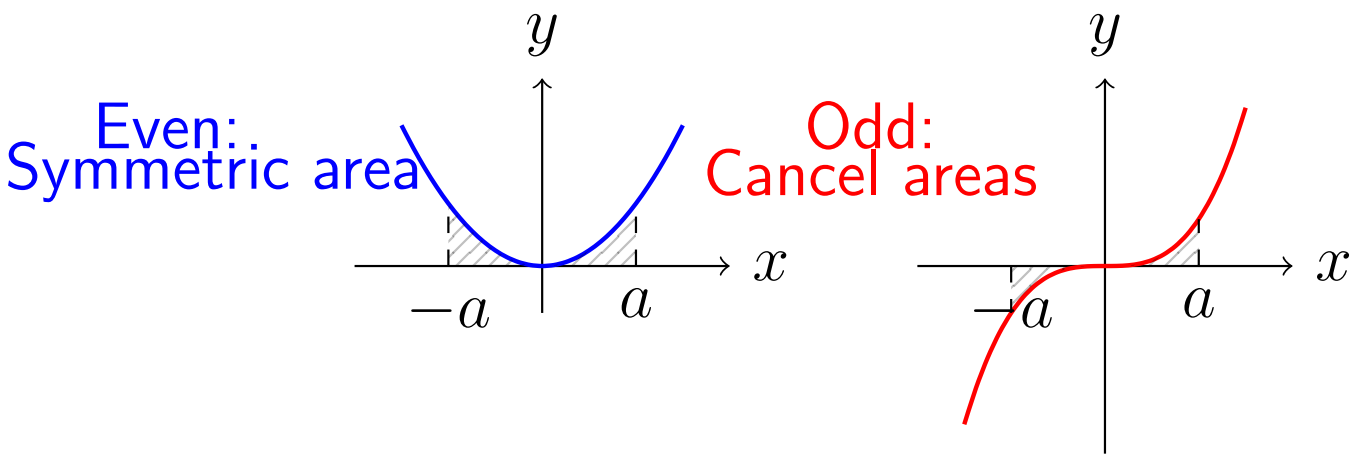
Even and Odd Function Property

Property (Even Functions)

If f *is* **even** $[f(-x) = f(x)]$, *then* $\int_{-a}^{a} f(x)dx = 2\int_{0}^{a} f(x)dx$.

Property (Odd Functions)

If f *is* **odd** $[f(-x) = -f(x)]$, *then* $\int_{-a}^{a} f(x)dx = 0$.



Worked Example: Even/Odd Property

Example

Evaluate $\int_{-2}^{2}(3x^5 - 2x^3 + 7x)dx$ without finding an antiderivative.

Solution: Check if the integrand is even or odd.
Let $f(x) = 3x^5 - 2x^3 + 7x$.

$$f(-x) = 3(-x)^5 - 2(-x)^3 + 7(-x) = -3x^5 + 2x^3 - 7x = -f(x).$$

So $f(x)$ is odd. The limits are symmetric about 0 (-2 to 2).
Therefore:

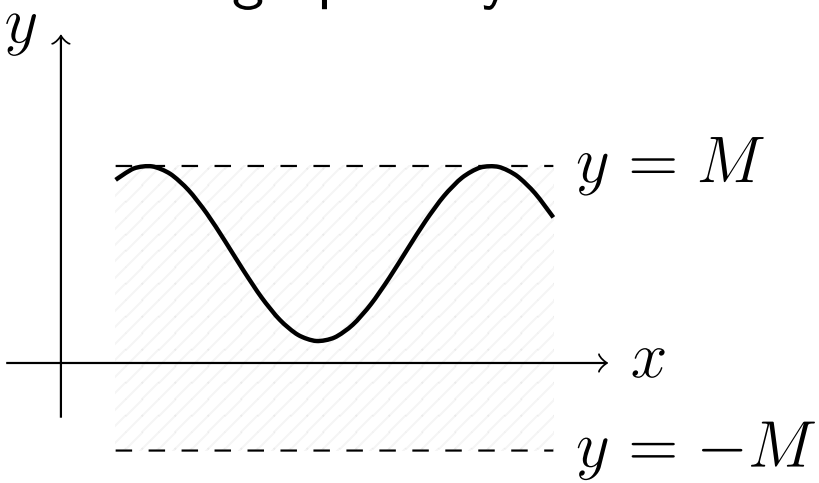
$$\int_{-2}^{2}(3x^5 - 2x^3 + 7x)dx = 0.$$

Bounded Functions

A function f that is defined on an interval is said to be **bounded** on the interval if there is a positive number M such that

$$|f(x)| \leq M \quad \text{for all } x \text{ in the interval.}$$

$|f(x)| \leq M$ means the graph stays between these lines



Simple way to think about it: A bounded function doesn't "blow up" to infinity—it stays within some fixed horizontal strip.

Integrability of Functions with Discontinuities

Theorem (Integrability of Bounded Functions with Finite Discontinuities)

Let f be a function that is defined on the finite closed interval $[a, b]$.

1. *If f has* **finitely many discontinuities** *in $[a, b]$ but is* **bounded** *on $[a, b]$, then f is integrable on $[a, b]$.*
2. *If f is* **not bounded** *on $[a, b]$, then f is* **not integrable** *on $[a, b]$.*

Integrability of Functions with Discontinuities

Theorem (Integrability of Bounded Functions with Finite Discontinuities)

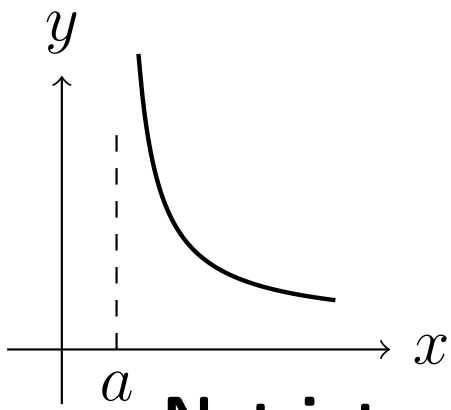
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What This Means in Practice

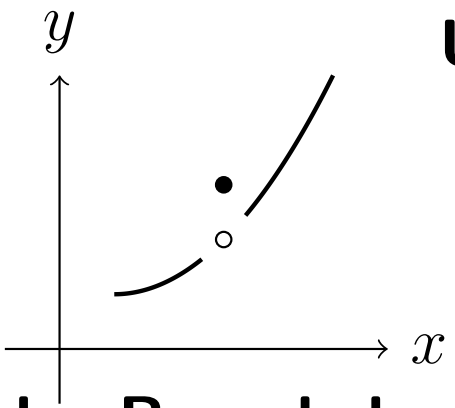
- **Bounded + finitely many "holes" or "jumps"** = Integrable
- **Unbounded (goes to $\pm\infty$)** = Not integrable

Visual Examples



Not integrable:

Unbounded near a



Integrable: Bounded, one hole

Worked Example

Determine if $f(x)$ is integrable on $[0, 3]$:

$$f(x) = \begin{cases} x^2 & \text{if } 0 \leq x < 1 \\ 2 & \text{if } x = 1 \\ x + 1 & \text{if } 1 < x \leq 2 \\ 4 - x & \text{if } 2 < x \leq 3 \end{cases}$$

1. **Check for discontinuities:**
   - At $x = 1$: $\lim_{x \to 1^-} f(x) = 1$, $\lim_{x \to 1^+} f(x) = 2$, $f(1) = 2 \Rightarrow$ jump discontinuity
   - At $x = 2$: $\lim_{x \to 2^-} f(x) = 3$, $\lim_{x \to 2^+} f(x) = 2$, $f(2) = 3$ (from left) $\Rightarrow$ jump discontinuity

   So f has 2 discontinuities (finitely many).
2. **Check boundedness:** On $[0, 3]$, $f(x)$ is between 0 and 4. Choose $M = 5$, then $|f(x)| \leq 5$ for all $x \in [0, 3]$. So f is bounded.

Worked Example: Non-Integrable Function

Example

Is $g(x) = \dfrac{1}{\sqrt{x-1}}$ integrable on $[1, 4]$?

- The function has a potential issue at $x = 1$ because the denominator becomes 0.
- As $x \to 1^+$, $\sqrt{x-1} \to 0^+$, so $g(x) = \frac{1}{\sqrt{x-1}} \to +\infty$.
- Therefore, g is **unbounded** on $[1, 4]$.
- By the theorem (part 2), if f is not bounded on $[a, b]$, then f is not integrable on $[a, b]$.
- Conclusion: g is **not integrable** on $[1, 4]$.

Practice Problems: Boundedness and Integrability

Practice Problem

Is the function $f(x) = \dfrac{1}{x-2}$ bounded on $[1,3]$? Is it integrable on $[1,3]$?

Practice Problem

True or False: A function with infinitely many jump discontinuities on $[a,b]$ cannot be integrable.

Practice Problem

Find a value of M such that $|f(x)| \leq M$ for $f(x) = x^2 - 4x + 7$ on $[0,3]$.

Key Points to Remember

- **Bounded function:** Stays between $-M$ and M for some $M > 0$.
- **Integrability theorem:**
  - Finitely many discontinuities + bounded = Integrable
  - Unbounded = Not integrable
- **Important:** A function can have discontinuities and still be integrable, as long as it's bounded and has only finitely many jumps/holes.
- **Warning:** If a function goes to $\pm\infty$ anywhere in $[a, b]$, it's not integrable on that interval.